AI 500 - Practice Problem Set 5

Q1. A
Q2. B, C
Q3. A
Q4. C
Q5. 5
Q6. (65 − 40) − (70 − 50) = 25 − 20 = 5.
Q7a)

Yes — in the general population, the presence of diabetes and gallbladder disease are independent events. Hence, no correlation exists.

Q7b)

Among hospitalized patients, the two diseases appear negatively correlated:
If a hospitalized patient doesn't have diabetes, they are more likely to have gallbladder disease (since something must have caused the admission).
Conversely, if they do have diabetes, they are less likely to also have gallbladder disease — because either condition alone suffices for hospitalization.
So within the hospitalized subset, one disease "compensates" for the absence of the other, creating a spurious negative correlation.

Q7c)

This occurs because hospitalization acts as a collider — a common effect of both diseases. When we condition on "being hospitalized," we open a non-causal path between Diabetes and Gallbladder Disease: D → H ← G

Q8a) ATE = 0.78 - 0.83 = -0.05
Q8b) Men

- Treated: 81/87 = 0.9310
- Control: 234/270 = 0.8667

CATE = 0.9310 - 0.8667 = 0.0643

Women

- Treated: 192/263 = 0.7308
- Control: 55/80 = 0.6875

CATE = 0.7308 - 0.6875 = 0.0433

Q8c) 0.0538

Q8d) Despite positive effects within both men and women, the overall difference is negative. This happens because the treated group has many more women (who, regardless of treatment, have lower recovery than men in this dataset). Aggregating without accounting for gender overweights the lower-baseline subgroup in the treated group, pulling down the overall treated rate.

Q9a) Yes, the effect of T on Y is identifiable through Z.

Q9b) Control for Z.

Q9c)

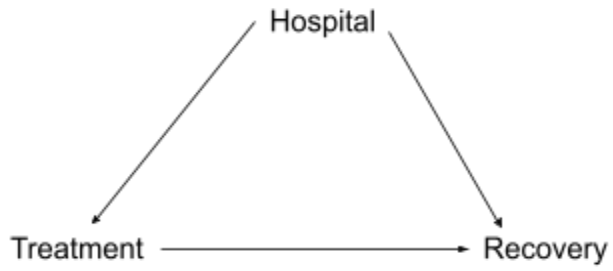
$$\text{ATE} = E[Y \mid do(T=1)] - E[Y \mid do(T=0)] = \sum_z \big(E[Y \mid T=1, Z=z] - E[Y \mid T=0, Z=z]\big) P(Z=z)$$

Q10a) This is a classic Simpson's paradox situation. The per-hospital summary numbers hide the fact that the two hospitals treat different mixes of patients. If Hospital A treats many low-risk patients who are easy to recover, and Hospital B treats many high-risk patients or vice-versa, then when you pool without adjusting for the hospital, the weighted average can reverse the apparent ordering.

In short: different group sizes and different treatment assignment rates across hospitals can flip the marginal (pooled) comparison relative to the within-hospital comparisons.

Q10b) Hospital is the confounder.

Q10c)



Q10d) Aggregating over groups without adjusting for group differences can produce misleading conclusions (reversal). You should condition on the confounding variable (here: hospital or patient severity) before comparing outcomes.

Q11a) Compute conditional probabilities from the admitted sample:

- Number with asthma = 10+40=5010 + 40 = 5010+40=50.
  (Pneumonia∣Asthma)=10/50=0.20.

- Number without asthma = 40+10=5040 + 10 = 5040+10=50.
  P(Pneumonia∣No Asthma)=40/50=0.80.

So in the admitted sample, asthma and pneumonia appear negatively related (people with asthma are less likely to have pneumonia in this admitted sample).

Q11b) they are independent.

Q11c) Admission is a selection rule that depends on the two conditions: patients with at least one condition are more likely to be in the admitted sample. That selection makes people who have one condition (say asthma) more likely to *not* have the other in the admitted set, producing an apparent negative correlation even though the two conditions are independent in the population. In plain terms: restricting the dataset to a selected subgroup (admitted patients) creates a misleading negative association — a selection artifact — between two variables that are independent in the full population.